ASSET PRICING

Modern Portfolio Theory & Factor Models

01 **CAPM**

02 **Fama-French 3-Factor Model**

03 Efficient Frontier

04 PCA in Finance

05 de Prado's HRP

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01 Capital Asset Pricing Model (CAPM)

Sharpe (1964) • Lintner (1965) • Mossin (1966)

CAPM FORMULA

$$E[R_i] = R_f + \beta_i \times \underbrace{(E[R_m] - R_f)}$$ ← Market Risk Premium

NOTATION

- $E[R_i]$ Expected return of asset i
- R_f Risk-free rate (T-bill)
- β_i $Cov(R_i, R_m) / Var(R_m)$
- MRP $E[R_m] - R_f$

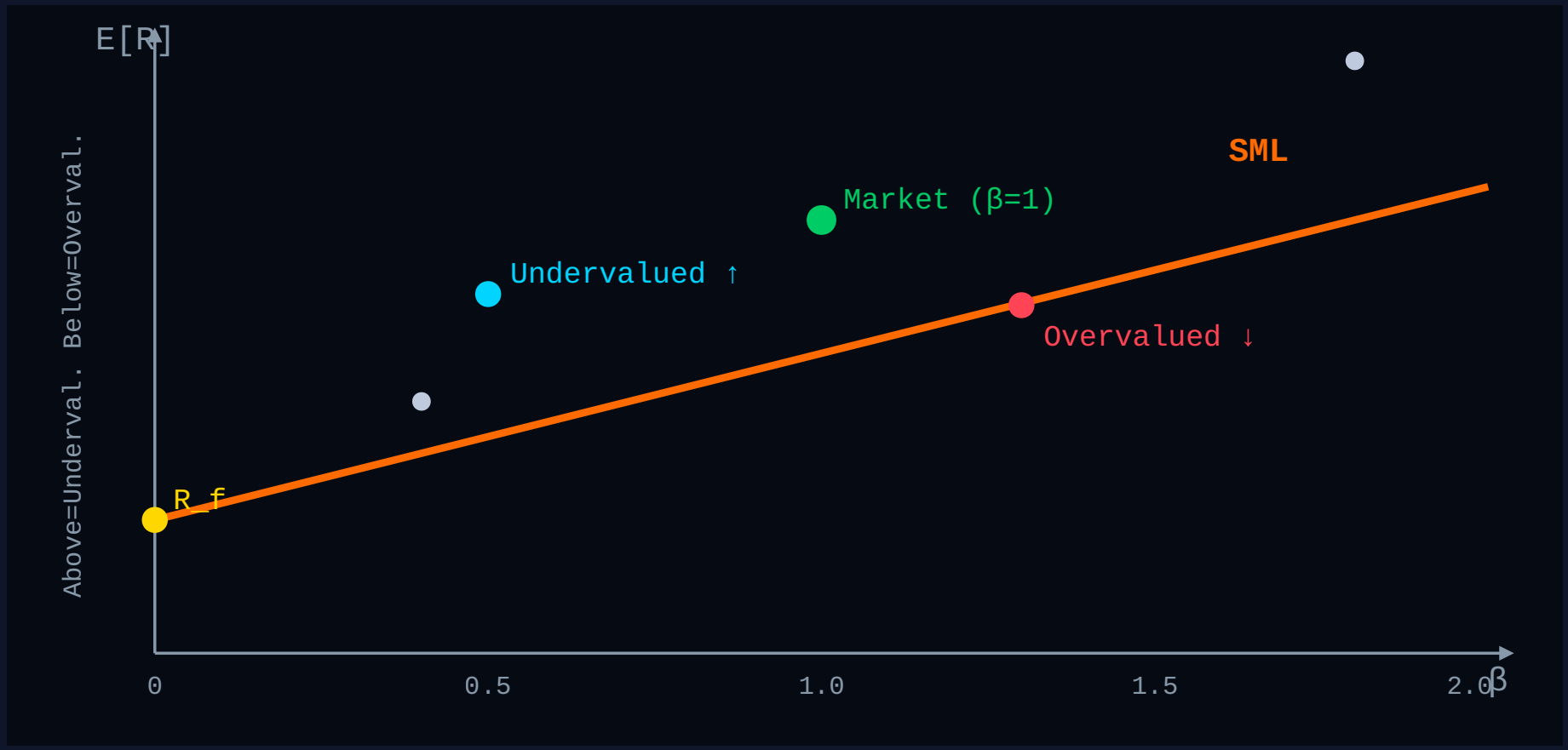
BETA INTERPRETATION

- $\beta < 1$ Defensive
- $\beta = 1$ Tracks market
- $\beta > 1$ Aggressive
- $\beta < 0$ Hedge asset

KEY ASSUMPTIONS

- Single-period horizon
- Homogeneous expectations
- No taxes / transaction costs
- Mean-variance utility
- Unlimited short-selling

SECURITY MARKET LINE (SML)



02 Fama-French 3-Factor Model

Fama & French (1992, 1993) — Beyond Beta: Capturing Size & Value Anomalies

FF3 FORMULA

$$E[R_i] - R_f = \alpha_i + b_i \cdot MKT + s_i \cdot SMB + h_i \cdot HML + \varepsilon_i$$

MKT
Market Factor

Construct: $R_m - R_f$
Market excess return; same as CAPM β exposure.

Premium: ≈ 5–8% p.a.

SMB
Small Minus Big

Construct: Small-cap − Large-cap
Small-cap historically outperforms large-cap.

Premium: ≈ 2–4% p.a.

HML
High Minus Low

Construct: High B/M − Low B/M
Value stocks (high B/M) outperform growth.

Premium: ≈ 3–5% p.a.

Metric	CAPM	FF3	Improvement
R^2 (avg)	≈ 70%	≈ 90%	+20 pp
Avg α	Significant	Near zero	✓ Captured
Size premium	Not explained	SMB factor	✓ Modeled
Value premium	Not explained	HML factor	✓ Modeled
Momentum	Not explained	Not explained	✗ Missing → Carhart 4F

Extensions: Carhart 4F (+Momentum, 1997) • FF5 (+RMW Profitability +CMA Investment, 2015) • q-factor (Hou et al.)

03 Efficient Frontier & Modern Portfolio Theory

Harry Markowitz (1952) — Mean-Variance Optimization

OPTIMIZATION PROBLEM

$$\min_w \; w^T \Sigma w$$

$$\text{s.t.} \quad w^T \mu = \mu^* \;|\; w^T 1 = 1 \;|\; w \geq 0$$

KEY METRICS

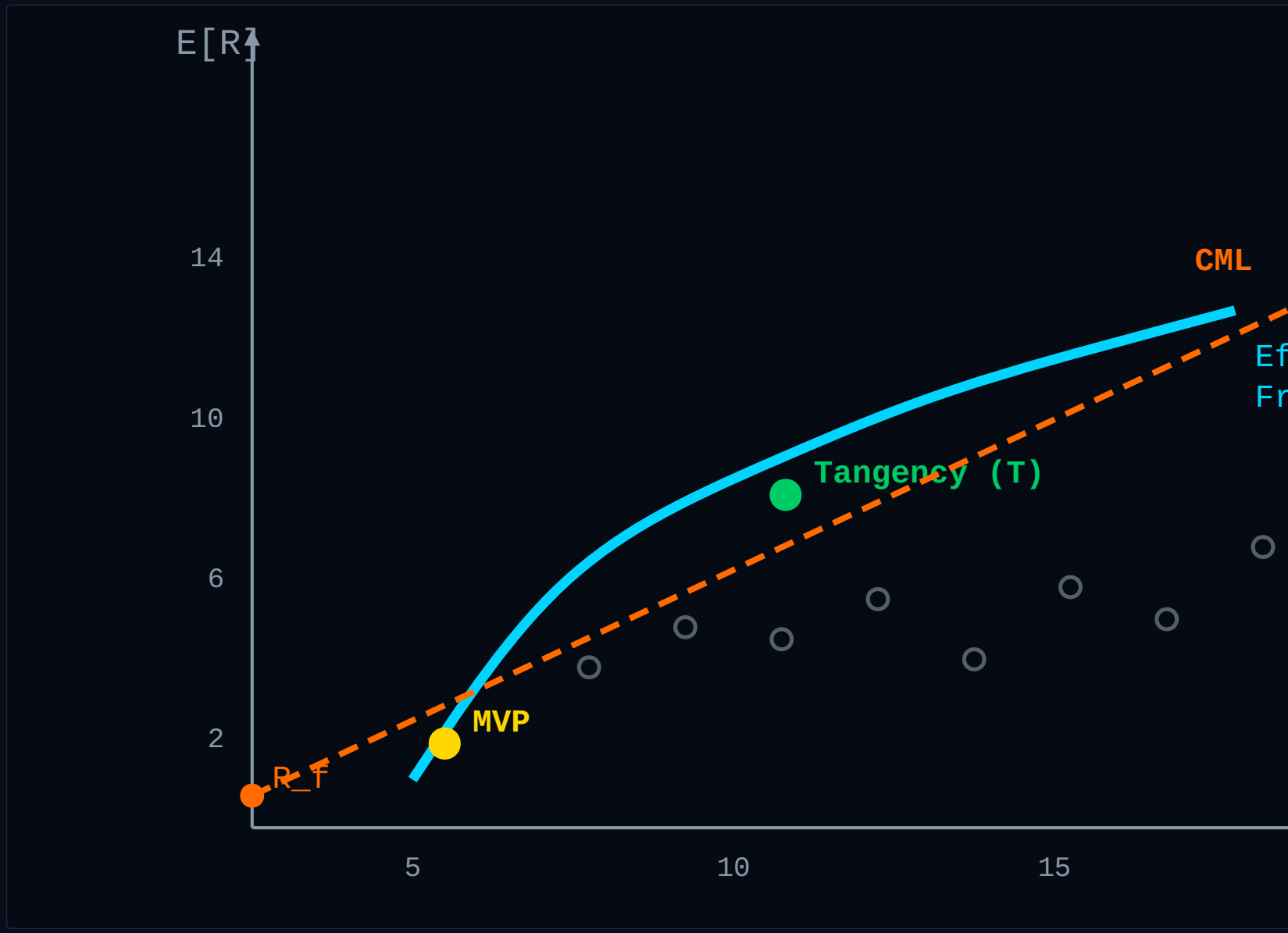
Sharpe	$(E[R_p] - R_f) / \sigma_p$
Port. Var.	$w^T \Sigma w$
Correlation	$\rho_{ij} = \sigma_{ij} / (\sigma_i \sigma_j)$
Diversif.	$\sigma_p < \sum w_i \sigma_i$ when $\rho < 1$

```
from pypfopt import (
    EfficientFrontier, risk_models,
    expected_returns)
mu = expected_returns.mean_historical_return(px)
S  = risk_models.sample_cov(px)
ef = EfficientFrontier(mu, S)
w  = ef.max_sharpe() # tangency portfolio
```

MVP
Min variance regardless of return

TANGENCY
Max Sharpe; optimal risky portfolio

CML
R_f+Tangency combos



04 Principal Component Analysis in Finance

Dimensionality Reduction • Latent Factor Discovery • Risk Decomposition

MATH FOUNDATION

$X = USV^T$ (SVD)

$\Sigma = Q\Lambda Q^T$ (Eigendecomp)

$PC_k = X \cdot v_k$

NOTATION

- Σ Covariance matrix (p×p)
- Q Eigenvectors (loadings)
- Λ Diagonal eigenvalue matrix
- v_k k-th PC direction
- λ_+ M-P noise threshold

MARCHENKO-PASTUR

$$\lambda_+ = \sigma^2(1 + \sqrt{(p/T)})^2$$

Eigenvalues $\leq \lambda_+$ are random noise

RISK MODEL

Low-rank Σ approx. reduces estimation noise

FACTOR DISCOVERY

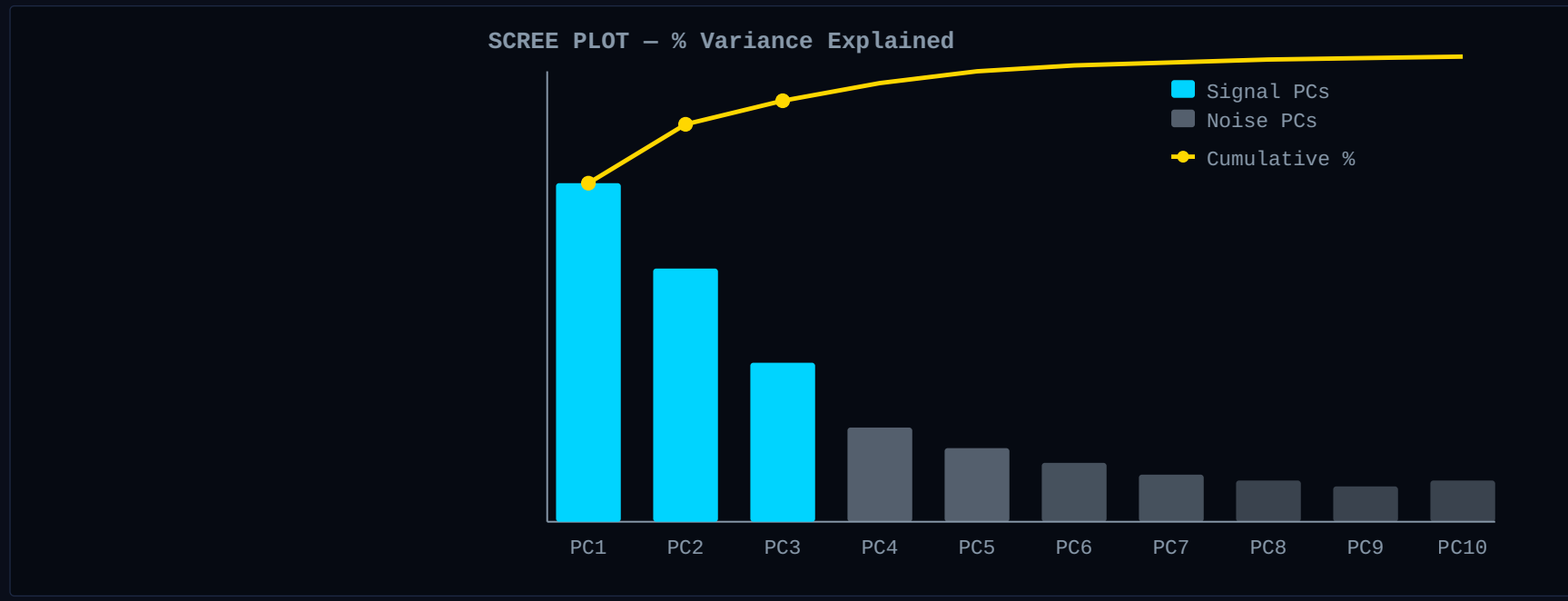
PC1≈Market, PC2≈Size, PC3≈Value

NOISE FILTER

Discard eigenvalues below λ_+ (M-P law)

DECORRELATION

Rotate to uncorrelated PC space before optimisation



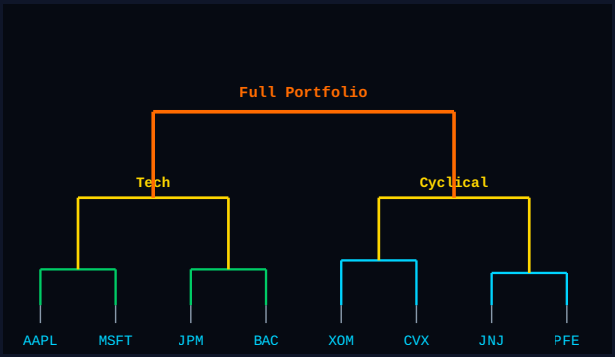
PCA PIPELINE

Returns Matrix X → Standardise (z-score) → Covariance $\Sigma = X^TX/T$ → Eigendecomp $\Sigma = Q\Lambda Q^T$ → Select Top-k PCs → Factor Portfolios

05 Hierarchical Risk Parity (de Prado)

Marcos Lopez de Prado (2016) — ML for Asset Managers • No matrix inversion required

HIERARCHICAL CLUSTERING DENDROGRAM



3-STEP ALGORITHM

STEP 1 **Tree Clustering:** Compute corr. matrix → distance $d_{ij}=\sqrt{(1-\rho_{ij})/2}$ → Ward linkage

STEP 2 **Quasi-Diagonalisation:** Reorder assets so correlated pairs are adjacent in the covariance matrix

STEP 3 **Recursive Bisection:** Split clusters; allocate inverse-variance weight within each cluster

Method	Main Weakness	Verdict
Inv-Variance	Ignores correlations	Naïve
Min-Variance	Noisy Σ^{-1}	Fragile
Equal Weight	Ignores risk structure	Simple
HRP	Robust + structure	✓ **Best**

```
from mlfinlab.portfolio_optimization import (
    HierarchicalRiskParity)
hrp = HierarchicalRiskParity()
hrp.allocate(asset_prices=prices, linkage='ward')
weights = hrp.weights  # pd.DataFrame
```

Ref: de Prado, M.L. (2016). Building diversified portfolios that outperform out of sample. *J. Portfolio Management* 42(4), 59–69.

DISTANCE METRIC

$$d_{ij} = \sqrt{(1-\rho_{ij}) / 2} \in [0,1]$$

SUM Asset Pricing — Summary & Connections

From CAPM to Machine Learning Portfolio Construction

Model	Year	Factors	Risk Measure	Key Contribution	Limitation
CAPM	1964	1 (Market)	Beta	First rigorous risk-return model	Unrealistic assumptions
FF3	1993	3 (MKT+SMB+HML)	Multi-β	Size & value anomalies explained	Momentum not captured
Eff. Frontier	1952	All assets (Σ,μ)	Variance	Formal portfolio optimisation	Sensitive to est. errors
PCA	(fin.)	k latent PCs	Eigenvalues	Data-driven latent factors	No econ. interpretation
HRP	2016	Hierarchical	Cluster Var.	No matrix inversion needed	Newer; less tested

- **CAPM** 1952–64
- **Markowitz** 1952
- **FF3** 1993
- **Carhart** 1997
- **FF5** 2015
- **HRP** 2016
- **ML/DL** Now

RISK DECOMPOSITION

CAPM: β
FF3: β, s, h
PCA: eigenvalues
HRP: cluster variance

PORTFOLIO CONSTRUCTION

MVO (Markowitz)
Risk Parity
Inverse-Variance
HRP (de Prado)

PYTHON ECOSYSTEM

`pypfopt`
`mlfinlab`
`sklearn / numpy`
`statsmodels`

WWAI APPLICATION

LRS signal + FF3
HRP allocation
PCA noise filter
GNN spillover map